Software Engineering & Project Management (SEPM)

# Energy & Utility Management System (EUMS)

Billing, metering and customer service for electricity, water and gas

## Use Cases and User Stories

Software process model: Waterfall · Combined SEPM + DBMS course project

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|---|---|
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# 1. Requirements Model: Use Cases and User Stories

## 1.1 Use-case diagram

Three actors use the system. "Employee" covers the meter reader, billing officer and technician, who share one role in the system. UC-05 *Generate bill* is never started by a person; it is included in UC-04, so every saved reading produces a bill.

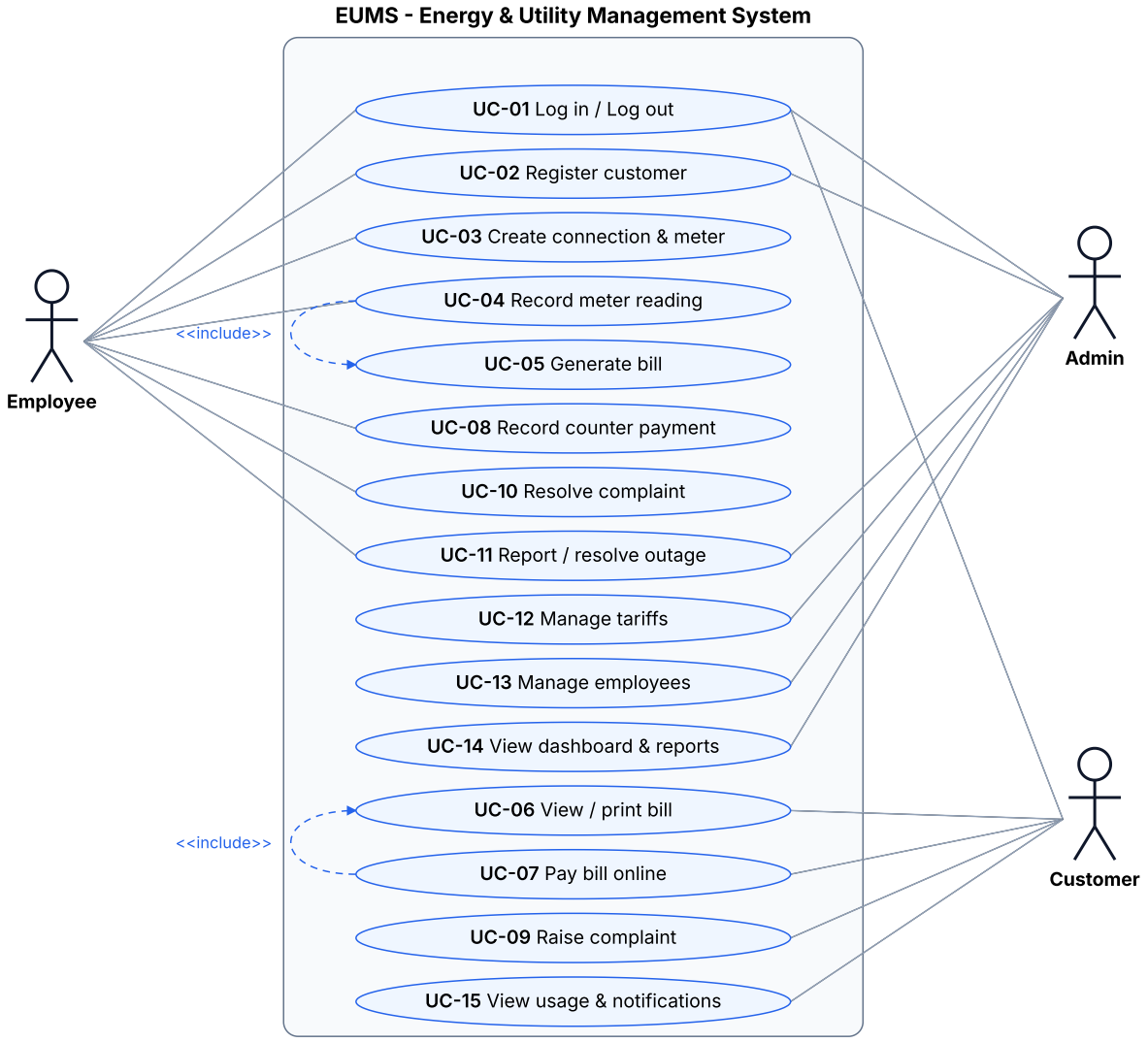



**Figure 1:** Use-case diagram of EUMS

## 1.2 Use-case descriptions

| UC-04 Record meter reading (includes UC-05 Generate bill) | |
|---|---|
| **Actor(s)** | Employee |
| **Pre-condition** | Employee is logged in; the connection and meter are active. |

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|---|---|
| **Main flow** | 1. Employee opens Meter Readings → Add Reading.<br>2. Employee selects a meter; system shows customer name, utility and previous reading with its date.<br>3. Employee types the current reading; system shows the consumption immediately.<br>4. Employee enters the reading date and optional remarks and clicks Save Reading.<br>5. System saves the reading, calculates the bill from the tariff slabs (UC-05), saves the bill and its items, and notifies the customer.<br>6. System opens the new bill. |
| **Alternate / exception flows** | • 3a. Current reading lower than previous: an error is shown and Save is disabled. If submitted anyway, the database trigger rejects it and nothing is saved.<br>• 4a. Date on or before the last reading date: the trigger rejects it and an error is shown.<br>• 4b. Reading already exists for that date: error shown. |
| **Post-condition** | Reading and bill stored; customer sees the bill and a "New bill generated" notification. |

| **UC-07 Pay bill online** | |
|---|---|
| **Actor(s)** | Customer |
| **Pre-condition** | Customer is logged in and has a bill that is unpaid or partial. |
| **Main flow** | 1. Customer opens Bills & Payments and clicks Pay on a bill.<br>2. System shows bill details and the outstanding amount, pre-filled.<br>3. Customer optionally changes the amount (partial payment) and chooses card / UPI / net banking / wallet.<br>4. Customer clicks Pay.<br>5. System records the payment with a receipt number; trigger updates bill to paid or partial and notifies the customer.<br>6. System shows the bill with the payment and a success message. |
| **Alternate / exception flows** | • 1a. Bill belongs to another customer: HTTP 403.<br>• 4a. Amount more than outstanding: trigger rejects; error shown; nothing saved.<br>• 4b. No method chosen: error shown.<br>• 1b. Bill already paid: message "already paid". |
| **Post-condition** | Payment saved; bill status updated; receipt number shown. |

| **UC-09 / UC-10 Raise and resolve complaint** | |
|---|---|
| **Actor(s)** | Customer, Employee |
| **Pre-condition** | Users are logged in. |
| **Main flow** | 1. Customer opens Complaints, chooses connection, types the issue, details and urgency, and submits.<br>2. System creates complaint CMPxxxx with status Open.<br>3. Employee opens Complaints, filters Open, and opens the complaint.<br>4. Employee assigns a technician, sets priority, adds notes and changes status to In Progress or Resolved.<br>5. System saves the change, records resolution time, and notifies the customer through a trigger. |
| **Alternate / exception flows** | • 1a. Issue left empty: error shown. |

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| **Post-condition** | Complaint tracked to resolution; customer informed of each status change. |

| UC-11 Report outage | |
|---|---|
| **Actor(s)** | Admin, Employee |
| **Pre-condition** | User is logged in as staff. |
| **Main flow** | 1. Staff opens Outages → Report Outage.<br>2. Staff selects utility and area and enters reason, start time and estimated end.<br>3. System saves the outage as Active (or Scheduled if it starts in the future).<br>4. Trigger notifies every customer in that area who has that utility.<br>5. Later, staff clicks Mark resolved. |
| **Alternate / exception flows** | • 2a. End time not after start time: error shown. |
| **Post-condition** | Outage visible to staff and affected customers. |

| UC-01 Log in | |
|---|---|
| **Actor(s)** | All |
| **Pre-condition** | User has an active account. |
| **Main flow** | 1. User opens the site and enters username/email and password.<br>2. System checks the password hash.<br>3. System opens the dashboard for the user's role. |
| **Alternate / exception flows** | • 2a. Wrong credentials: "Invalid username or password".<br>• Any page opened without login redirects to the login page. |
| **Post-condition** | User session created. |

## 1.3 User stories with acceptance criteria

| ID | User story | Acceptance criteria | Status |
|---|---|---|---|
| US-01 | As a **meter reader**, I want to see the previous reading when I pick a meter, so that I do not mistype the new one. | Given a meter is selected, previous value and date are shown; consumption updates while typing; lower values show an error and disable Save. | Done |
| US-02 | As a **billing officer**, I want bills to be calculated automatically from slabs, so that there are no manual errors. | For 250 kWh on the domestic plan the bill is ₹1,207.50 with two slab rows, fixed charge and 5% tax. | Done |
| US-03 | As a **customer**, I want to pay my bill online by UPI or card, so that I do not have to visit the office. | Pay page shows outstanding amount; payment creates receipt RCPyyyymmdd-nnnn; bill becomes Paid; overpayment rejected. | Done |
| US-04 | As a **customer**, I want to see all my connections and current bills on one dashboard. | Dashboard shows one card per connection with latest bill amount, due date and status, plus a 6-month chart. | Done |
| US-05 | As a **customer**, I want to be told about outages in my area. | When staff report an outage for area A and utility U, all customers in A with U get a notification; others do not. | Done |
| US-06 | As a **technician**, I want to see open complaints and update them. | Complaints page has tabs All/Open/In progress/Resolved with counts; saving Resolved records the time and notifies the customer. | Done |
| US-07 | As an **admin**, I want to change tariff rates without changing old bills. | Edited rates are used for the next bill; previously issued bill totals are unchanged. | Done |

| | | | |
|---|---|---|---|
| US-08 | As an **admin**, I want to see revenue, dues and complaints at a glance. | Dashboard shows 6 KPI cards, billed amount by utility for 6 months, collections split and recent activity. | Done |
| US-09 | As an **admin**, I want overdue reminders and late fees to be applied automatically. | Bills unpaid after the due date get a surcharge and the customer is reminded. | Planned |
| US-10 | As a **billing officer**, I want to replace a faulty meter without breaking reading history. | Old meter marked Replaced; new meter starts at its own baseline; next bill uses the new meter. | Planned |

**Table 1:** User stories